

Cambridge International AS & A Level

MATHEMATICS**9709/13**

Paper 1 Pure Mathematics 1

May/June 2025**MARK SCHEME**

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **27** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
	Ignore subsequent work
	Judgement
	Judgement
	Method mark awarded zero
	Method mark awarded one
	Method mark awarded two
	Misread
	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Judgment made by the PE
	Premature approximation
	Special case
	Indicates that work/page has been seen

Annotation	Meaning
	Error in number of significant figures
	Correct
	Transcription error
	Correct answer from incorrect working

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

Question	Answer	Marks	Guidance
1	$2 + (12)(-2)x^{-3}$	B1	Correct differential but can be unsimplified.
	$2 + (12)(-2)(-2)^{-3} [= 5]$	*M1	Substitute $x = -2$ into <i>their</i> differential, which must contain x^{-3} .
	Either $(\text{their } 5) = \frac{y - (-1)}{x - (-2)}$ or $-1 = (\text{their } 5) \times (-2) + c \Rightarrow c =$	DM1	Attempt to find equation of tangent through $(-2, -1)$ with their numerical gradient obtained as described above.
	$y = 5x + 9$	A1	
		4	

Question	Answer	Marks	Guidance
2	$r = \frac{8 \sin^3 \theta}{4 \sin^2 \theta} [= 2 \sin \theta]$	B1	A correct unsimplified expression for r , this may only be seen in a correct S_{∞} .
	$\frac{4 \sin^2 \theta}{1 - 2 \sin \theta} = \frac{1}{2}$	*M1	OE Correct S_{∞} with <i>their</i> expression for r .
	Or		
	$r = \frac{8 \sin^3 \theta}{4 \sin^2 \theta}$ and $\frac{4 \sin^2 \theta}{1 - r} = \frac{1}{2}$	B1	
	$1 - 8 \sin^2 \theta = \frac{8 \sin^3 \theta}{4 \sin^2 \theta}$	*M1	
	Then		
	Expect $8 \sin^2 \theta + 2 \sin \theta - 1 [= 0]$	DM1	OE and attempt to solve. Attempt to solve a three-term quadratic to find at least one value of $\sin \theta$. This can be implied by a final answer of $\theta = 0.253^{\circ}$ AWRT
	$[\theta =] \sin^{-1} \frac{1}{4}, 0.25$	A1	OE Clearly identify one answer only. Accept $\sin \theta = \frac{1}{4} \Rightarrow k = \frac{1}{4}$ etc.
		4	

Question	Answer	Marks	Guidance
3	$\left(\frac{1}{-1 \times 4}\right)a(4x-3)^{-1} + 2x$	B1	OE Do not accept $(-2+1)$ as equivalent to -1 .
	Apply correct limits, $x = 3$ & 1 , to <i>their</i> integral	*M1	<i>Their</i> integral must contain $(4x-3)^{-1}$. Condone using $x = 1$ and 3 .
	$\frac{-a}{36} + 6 - \left(\frac{-a}{4} + 2\right) = 12 \Rightarrow \left[\frac{8a}{36} + 4 = 12\right]$	DM1	OE Equate <i>their</i> linear unsimplified expression in a to 12 .
	$a = 36$	A1	
		4	

Question	Answer	Marks	Guidance
4(a)	$\{32[x^0]\}\{-120x\}\{+180x^2\}$	B1B1B1	Coefficients must be simplified. Ignore further terms in the expansion. SC B3: $2^5\left(1 - \frac{15}{4}x + \frac{45}{8}x^2 \dots\right)$ can be awarded full marks, ISW.
		3	
4(b)	$x = 0.01$ or $\frac{1}{100}$	B1	Identify 0.01 as the only value to substitute, condone any method.
	$(\textit{Their } 32 - 120x + 180x^2)$ with $x = (\textit{their } 0.01)$	M1	Clear use of <i>their</i> x -value from <i>their</i> solution of $2 - \frac{3}{2}x = 1.985$ and <i>their</i> answer to (a). Expect $32 - 120(0.01) + 180(0.01)^2 = 32 - 1.2 + 0.018$. Condone substitution into further terms in the expansion.
	30.8[18]	A1	Correct answer to at least 3sf coming from substitution into three or more terms. 'Correct' answers without appropriate working get A0.
		3	

Question	Answer	Marks	Guidance
5	Replace $\tan \theta$ with $\frac{\sin \theta}{\cos \theta}$	M1	
	Replace $\sin^2 \theta$ with $1 - \cos^2 \theta$ leading to a 3-term quadratic	*M1	To obtain a 3-term quadratic in $\cos \theta$. Accept a cubic that has a common factor of $\cos \theta$. Condone +/- sign errors during simplification.
	$9 \cos^2 \theta + \cos \theta - 4 [= 0]$	A1	OE
	Attempt to solve (<i>their</i> quadratic in $\cos \theta$) using a valid method	DM1	Only available for solution of a three-term quadratic. Correct values of $\cos \theta$ are 0.6134 and -0.7245 .
	Any two of ± 52.2 , ± 136.4	A1	AWRT as final answers. Any two correct answers between -180° and 180° as <i>their</i> final answers.
	All four values	A1	Condone other answers outside the range but no others between -180° and 180° . SC after M1 *M1 A1 DM0, correct answers can score B1 B1.
		6	

Question	Answer	Marks	Guidance
6	$a + 2(N - 1) = 55$	B1	SOI
	$\frac{1}{2} \times 3N(2a + 2(3N - 1)) = 5760$	B1	SOI
	Eliminate either a or N	M1	Using a valid substitution, expect either $a = 57 - 2N$ or $N = \frac{(57 - a)}{2}$. Condone +/- sign errors in <i>their</i> substitution or during simplification.
	$N^2 + 56N - 1920$ or $a^2 - 226a + 1953 [= 0]$	A1	All terms need not be collected on one side.
	$N = 24, a = 9$	A1 A1	Ignore presence of extra solutions for N and a .
		6	

Question	Answer	Marks	Guidance
7(a)	$3x^2 + 10x - 8 < 0$ or $3\left(x + \frac{5}{3}\right)^2 - \frac{49}{3} < 0$	*B1	OE Condone = or ≤ 0 , but use of $>$ or $\geq$ can only score B0B1B0. If there is no $<$, = or ≤ 0 , then a correct final answer implies < 0 here.
	-4 and $\frac{2}{3}$	B1	
	$-4 < x < \frac{2}{3}$	DB1 FT	OE Use of $\leq$ sign(s) gets DB0. Condone $x > -4$, $x < \frac{2}{3}$, $x > -4$ and $x < \frac{2}{3}$ but not $x > -4$ or $x < \frac{2}{3}$. FT on <i>their</i> critical values.
		3	
7(b)	Identify $x = -4$ as the x -value of the maximum point.	B1 FT	SOI FT on <i>their</i> lower critical value from part (a).
	$[y =] \frac{3}{3}x^3 + \frac{10}{2}x^2 - 8x [+c]$	B1	Correct integration of the three terms given. Condone $f(x) =$.
	Use of $y = 27$, $x = \text{their}\left(-4 \text{ or } \frac{2}{3}\right)$ in <i>their</i> integral.	M1	<i>Their</i> integral must be a cubic, <i>their</i> $\left(-4 \text{ or } \frac{2}{3}\right)$ must come from (a) or a restart.
	$c = -21 \Rightarrow y = x^3 + 5x^2 - 8x - 21$	A1	Do not ISW if they continue to find a straight line. Condone omission of final statement if y or $f(x) = x^3 + 5x^2 - 8x + c$ has been seen earlier.
		4	

Question	Answer	Marks	Guidance
8(a)	$\frac{1}{2}\pi - 2\left(\tan^{-1}\frac{4}{12} \text{ or } \sin^{-1}\frac{4}{\sqrt{160}} \text{ or } \cos^{-1}\frac{12}{\sqrt{160}}\right)$ Or $2\sin^{-1}\left[\frac{\frac{1}{2}EF}{AE} = \frac{\sqrt{32}}{\sqrt{160}} = \frac{\sqrt{2}}{\sqrt{10}}\right] \text{ or other valid method.}$	M1	Attempt a complete valid method for finding angle EAF. $\sqrt{160}$ may be replaced by 12.65 AWRT and $\sqrt{32}$ by 5.657 AWRT. Working in degrees should give $E\hat{A}F = 53.13^\circ$ which converts to the correct value in radians.
	$[E\hat{A}F = 0.927295\dots] = 0.9273$	A1	AG If decimals are used, at least one of 0.32175..., 0.6435 or 0.92729.. must be seen in the first method and 5.6569 and 12.649 or 0.46364 seen in the second method for the correct level of accuracy. Note: $E\hat{A}F = 53.14^\circ$ does not convert to the correct answer, so A0.
	Alternative Method for Question 8(a)		
	$12^2 - (24 + 24 + 32) = \frac{1}{2}(their AE \times their AF) \sin E\hat{A}F$ $\sin E\hat{A}F = \frac{2\{12^2 - (24 + 24 + 32)\}}{(their AE \times their AF)}$ $E\hat{A}F = \sin^{-1} \frac{2\{12^2 - (24 + 24 + 32)\}}{(their AE \times their AF)}$	M1	Attempt a complete valid method for finding angle EAF. Area of ABCD – (Area Δs ADF, ABE & CEF) = Area ΔAEF. $E\hat{A}F = \sin^{-1}\left(\frac{4}{5} \right)$. Using the cosine rule should lead to $\cos^{-1} \frac{3}{5}$.
	$\left[E\hat{A}F = \sin^{-1} \frac{4}{5} = \right] \frac{\frac{1}{2}EF}{AE}$	A1	AG
		2	

Question	Answer	Marks	Guidance
8(b)	$[AE \text{ or } AF =] \sqrt{160} \text{ or } \frac{4}{\sin \hat{EAB}}$	B1	OE Expect AWRT 12.65.
	$[r\theta =] (their\ 12.65) \times 0.9273$	M1	Use of $r\theta$ with $(their\ \sqrt{160})$. Note: using $r = 12$ scores M0.
	$[11.729... + 8 + 8 =] 27.7$ AWRT	A1	
		3	
8(c)	$\frac{1}{2}(their\ 12.65)^2 \times 0.9273 [= 74.184]$	M1	Use $\frac{1}{2}r^2\theta$ for area of sector with $(their\ 12.65)$. Note: using $r = 12$ scores M0.
	$144 - 24 - 24 - \left\{ \frac{1}{2}(their\ 12.65)^2 \times 0.9273 \right\}$	M1	Attempt a complete method for finding area of shaded region. Condone use of $r = 12$.
	[Area =] 21.8	A1	AWRT
	Alternative Method for Question 8(c)		
	$\frac{1}{2}(their\ 12.65)^2 (0.9273 - \sin 0.9273) [= 10.183]$	M1	Area of the segment with $(their\ 12.65)$. Note: using $r = 12$ scores M0.
	$32 - \frac{1}{2}(their\ 12.65)^2 (0.9273 - \sin 0.9273)$	M1	Attempt a complete method for finding area of shaded region, condone use of $r = 12$.
	[Area =] 21.8	A1	AWRT
		3	

Question	Answer	Marks	Guidance
9(a)	Gradient of $PR = \frac{k-5}{2+13}$ or gradient of $RQ = \frac{k-1}{2-5}$	B1	Obtain at least one relevant gradient.
	$(\text{their gradient of } PR) \times (\text{their gradient of } RQ) = -1$ and attempt to simplify	M1	Clear of fractions and expand brackets. Expect $k^2 - 6k + 5 = 45$ OE. Condone +/- sign errors during simplification.
	$k^2 - 6k - 40 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	
	Alternative Method for Question 9(a)		
	$PR^2 = (k-5)^2 + (2+13)^2$ or $RQ^2 = (k-1)^2 + (2-5)^2$	B1	Obtain at least one of PR^2 and RQ^2 . These expressions may be seen under a square root sign.
	$(\text{their } PR^2) + (\text{their } RQ^2) = 18^2 + 4^2$ and attempt to simplify	M1	Expect $2k^2 - 12k + 260 = 340$ OE. Condone +/- sign errors.
	$2k^2 - 12k - 80 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	

Question	Answer	Marks	Guidance
9(a)	Alternative Method 2 for Question 9(a)		
	$x^2 + y^2 + 8x - 6y - 60 = 0$	B1	OE
	$2^2 + k^2 + 8 \times 2 - 6 \times k - 60 = 0$	M1	Substitution of $(2, k)$ into their circle equation.
	$k^2 - 6k - 40 [= 0]$	A1	OE
	$k = 10, k = -4$	A1	
		4	If none of the above marks are awarded, then SC B1 for using $k = 10$ to find the gradients of QR (-3) and PR ($\frac{1}{3}$) and showing that their product is -1 (or other similar verification), and then stating that this shows that 10 is a possible value for k .

Question	Answer	Marks	Guidance
9(b)	[Centre is] $(-4, 3)$	B1	SOI
	Radius gradient = $\frac{(their\ 3) - 10}{(their\ -4) - 2} \left[= \frac{7}{6} \right]$	*M1	Attempt at finding the gradient of the radius using $(2, 10)$ and their centre, but not P, Q , the mid-point of PR or QR .
	Either $\frac{-1}{\left(their\ \frac{7}{6} \right)} = \frac{y-10}{x-2}$ Or $10 = \frac{-1}{\left(their\ \frac{7}{6} \right)} \times 2 + c \Rightarrow c = \dots$	DM1	OE Correct method to find the equation of the tangent using $\frac{-1}{their\ radius\ gradient}$ and $(2, 10)$.
	$y - 10 = -\frac{6}{7}(x - 2)$ or $y = \left(-\frac{6}{7} \right)x + \frac{82}{7}$	A1	OE Correct equation but not stated in the required form.
	$6x + 7y - 82 = 0$ or $82 - 6x - 7y = 0$	A1	All correct terms on one side, but condone them being in the wrong order.
	Alternative Method for Question 9(b)		
	$x^2 + y^2 + 8x - 6y - 60 = 0$	B1	OE Equation of the circle.
	$2x + 2y \frac{dy}{dx} + 8 - 6 \frac{dy}{dx} = 0$ Or $\frac{dy}{dx} = \frac{-2x - 8}{2} (85 - x^2 - 8x - 16)^{-\frac{1}{2}}$	*M1	Differentiate implicitly to arrive at an expression with two terms containing $\frac{dy}{dx}$. Or rearrange to make y the subject and differentiate to arrive at an expression of the form $f'(x) \times f(x)$.

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Question	Answer	Marks	Guidance
9(b)	$2(2) + 2(10)\frac{dy}{dx} + 8 - 6\frac{dy}{dx} = 0$ Or $\frac{dy}{dx} = \frac{-2(2) - 8}{2} (85 - (2)^2 - 8(2) - 16)^{-\frac{1}{2}}$	DM1	Substitute (2,10) into their implicit differential. Or substitute $x = 2$ into their expression for $\frac{dy}{dx}$.
	$y - 10 = -\frac{6}{7}(x - 2)$ or $y = -\frac{6}{7}x + \frac{82}{7}$	A1	OE Correct equation but not stated in the required form.
	$6x + 7y - 82 = 0$	A1	All correct terms on one side, but condone them being in the wrong order.
		5	

Question	Answer	Marks	Guidance
10(a)	$-9 \times 2(2x-5)^{-2} + 2$	B1	Correct differential.
	$(\text{their } -18(2x-5)^{-2} + 2) = 0$ and rearrange to form a quadratic. $\left[(2x-5)^2 = 9 \text{ or } 8x^2 - 40x + 32 = 0 \right]$	M1	Equating a two term $\frac{dy}{dx}$ to 0 and dealing correctly with the negative power. Their two term $\frac{dy}{dx}$ must contain $(2x-5)^{-2}$.
	$(1, -6)$ and $(4, 6)$	A1, A1	A1 for two correct x -values or one correct point, second A1 for all correct.
		4	
10(b)	$-18 \times -2 \times 2(2x-5)^{-3} \left[= 72(2x-5)^{-3} \text{ or } \frac{144x-360}{(2x-5)^4} \right]$	B1 FT	Following through on <i>their</i> first derivative which must contain $(2x-5)^{-2}$.
	Use $(\text{their } x = 1 \text{ and } x = 4)$ in $(\text{their } 72(2x-5)^{-3})$ To determine the nature of both turning points.	M1	Substitute x -coordinate of each stationary point and determine their nature. Nature of the turning points must correctly follow from <i>their</i> values of $\frac{d^2y}{dx^2}$.
	For $x = 1$, $\left[\frac{d^2y}{dx^2} = \right] -\frac{72}{27}$ or $< 0 \Rightarrow$ maximum For $x = 4$, $\left[\frac{d^2y}{dx^2} = \right] \frac{72}{27}$ or $> 0 \Rightarrow$ minimum	A1	CWO
		3	

Question	Answer	Marks	Guidance
10(c)(i)	(1, -13)	B1	
		1	
10(c)(ii)	$[y =] \frac{9}{2(x \pm 3) - 5} + 2(x \pm 3) - 5 \pm 7$	M1	Application of $\begin{pmatrix} -3 \\ 7 \end{pmatrix}$ to the original expression for C but condone $+/-$ sign errors.
	$[y =] - \left(\frac{9}{2(x \pm 3) - 5} + 2(x \pm 3) - 5 \pm 7 \right)$	M1	SC B1 for $[y =] - \left(\frac{9}{2x - 5} + 2x - 5 \right)$.
	$y = -\frac{9}{2x + 1} - 2x - 8$	A1	Answer must be in this format; the ' $y =$ ' can be implied by earlier inclusion.
		3	

Question	Answer	Marks	Guidance
11(a)	$[f(x)](x+2a)^2 - 4a^2 + a$	B1	
	$(their(-4a^2 + a)) = -33$	M1	Condone $\geq$ or $>$.
	$4a^2 - a - 33 [= 0]$	B1	Condone $\leq$ or $<$.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.
	Alternative Method for Question 11(a)		
	$x = -2a, y = -4a^2 + a$	B1	The co-ordinates of the minimum point (likely to be either from differentiation or completing the square).
	$(their - 4a^2 + a) = -33$	M1	Their y-coordinate equated to -33 . Condone $\geq$ or $>$.
	$-4a^2 + a + 33 [= 0]$	B1	OE Condone $\geq$ or $>$.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.

Question	Answer	Marks	Guidance
11(a)	Alternative Method 2 for Question 11(a)		
	$x^2 + 4ax + a + 33 [=0]$	B1	Condone $\geq$ or $>$.
	$(4a)^2 - 4[1](a + 33) [=0]$	M1	Condone $\geq$ or $>$.
	$16a^2 - 4a - 132 [=0]$	B1	OE Condone $\geq$ or $>$. Accept $4a^2 - a - 33$ or multiples thereof.
	$-\frac{11}{4}, 3$	A1	OE Do not ISW if their final answer is given as a range or if one of the answers is rejected.
		4	

Question	Answer	Marks	Guidance
11(b)	$[g(x) =] \frac{1}{2}(x^3 + 4)$	B1	Expression for $g(x)$. SOI
	Either		
	$[g(0) =] \frac{1}{2}((0)^3 + 4) \quad [g(0) = 2]$	M1	Replacing x with 0 in <i>their</i> $g(x)$.
	$gg(0) = 6$	A1	
	$f(\text{their } 6) [= 36 + 24a + a]$	M1	
	Or		
	$fg(x) = \left(\frac{x^3 + 4}{2}\right)^2 + 4a\left(\frac{x^3 + 4}{2}\right) + a$ or $gg(x) = \frac{\left(\frac{x^3 + 4}{2}\right)^3 + 4}{2} \quad \left[= \frac{(x^3 + 4)^3}{16} + 2 \right]$	M1	Either composite function using <i>their</i> $g(x)$.
	$fgg(x) = \left\{ \frac{(x^3 + 4)^3}{16} + 2 \right\}^2 + 4a \left\{ \frac{(x^3 + 4)^3}{16} + 2 \right\} + a$	A1	Complete algebraic expression for $fgg(x)$.
	$fgg(0) = \left\{ \frac{(0 + 4)^3}{16} + 2 \right\}^2 + 4a \left\{ \frac{(0 + 4)^3}{16} + 2 \right\} + a$	M1	Replacing x with 0 in <i>their</i> $fgg(x)$.

Question	Answer	Marks	Guidance
11(b)	Then		
	$36 + 24a + a = 96$	A1	OE
	$[a =] \frac{12}{5}$	A1	OE
		6	